Sigma-style score temperature on CoBit

TinyGSM → GSM8K, executable-code accuracy · results of the experiments in “Testing Sigma-style score temperature on CoBit-S” (11 Oct 2026) · run 12–13 Oct 2026 on Isambard GH200 · CPU-only analysis · revised with Experiment 1 on the 500k checkpoint

Best fixed cell: A churn, unsharpened last step, τ=0.3 — 35.3% vs 28.1% for the original sampler at τ=1 (+7.1 [+5.1, +9.2] pts, paired) and 32.4% for the original best (τ=0.7). Details in Section 7.

Experiment 1 (unguided CoBit-S, 500k, EMA): 28.1% at τ=1 → **32.4% at τ=0.7** (+4.3 [+2.4, +6.2] pts, paired; churn, 1024 steps, entropic schedule). Global score temperature helps the main unguided model too; below τ=0.7 it collapses, most likely because the entropic σ grid's final step cannot take a drift that large (Section 5; fixed in Section 7), rather than because the model stops benefiting. **Across the two other checkpoints** it lifts CoBit-M (462M) from 31.5% to 39.3% (τ=0.3, +7.7 [+5.5, +9.9]) and the CoBit-S CFG checkpoint at w=1 from 25.9% to 31.9% (τ=0.2, +6.0 [+3.9, +8.1]). **CFG adds nothing on top** (best (w, τ) = (1, 0.2), 31.9%). **Sigma's τ ≤ 0.1 gives 0% with churn**: our EDM-churn Euler step is numerically unstable there, which says nothing about the readout.

Correction to the first version. It blamed a stale EMA for the unguided checkpoint's low scores. That was wrong: the EMA is complete and was updating normally (Section 4). Nor is it the σ schedule: the 500k EMA at τ=1 scores 28.1% (entropic) and 26.0% (Karras), reproducing the ~29% of June. The failure is specific to the uploaded 425k file the copy in the shared Drive folder is byte-identical to it (same SHA-256), so this is the Drive artifact itself; the June handoff records that the 29.42% came from a main-run 425k that was never saved to the shared folder; Experiment 1 now runs on the 500k checkpoint.

1 What was run

One GPU per cell, all 1319 GSM8K test problems, seed 0, EMA weights, DDIM/Euler integrator, `--score_temp_mode global`: the PF-ODE drift is divided by τ at every σ after the CFG mix, i.e. $D_\tau = x + (D - x)/\tau$, exactly Eq. (3) of the note. Self-conditioning and the final readout use the untempered posterior, as in Sigma. τ=1 is bit-identical to the previous sampler (unit-tested). Churn is EDM γ=0.41 per step over the whole σ range.

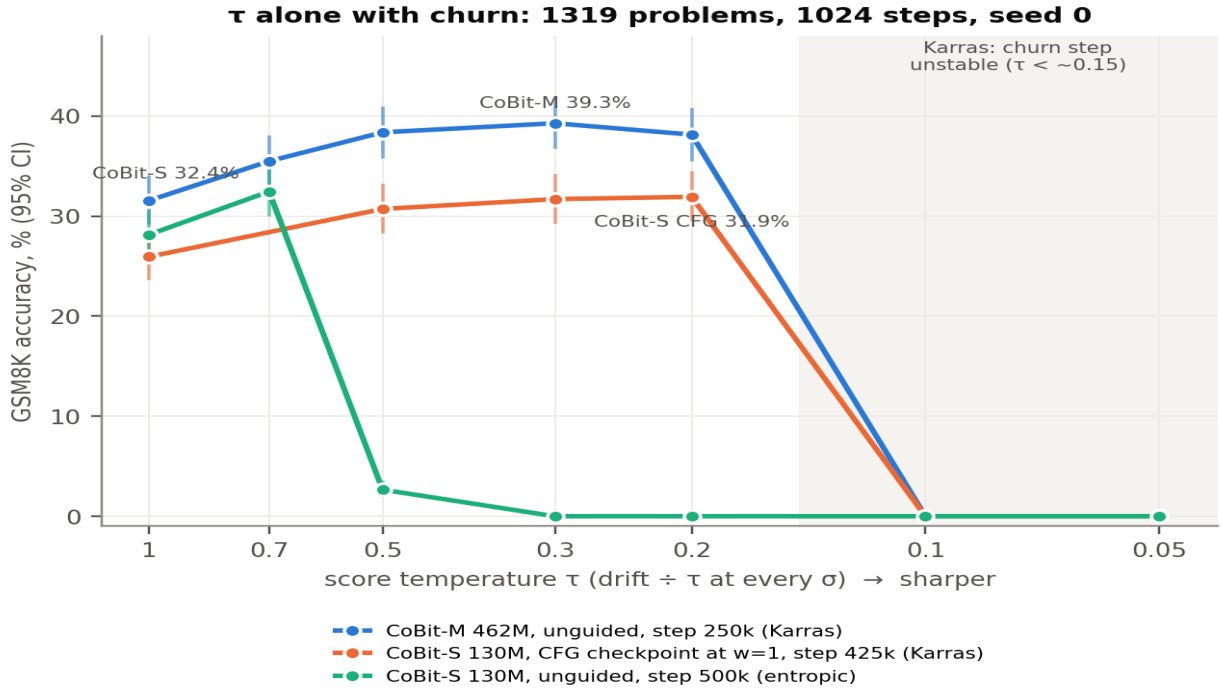
arm	checkpoint	σ schedule	cells
Exp. 1 — τ alone	CoBit-S unguided, step 500k (Drive last.pt)	entropic	τ ∈ {1, .7, .5, .3, .2, .1, .05} × {churn @1024, γ=0 @1024, churn @512}
schedule anchor	same, τ=1, churn @1024	entropic vs Karras	2
Exp. 2 — (w, τ)	CoBit-S p_{uncond}=0.1, step 425k	Karras	w ∈ {1, 2, 3, 5, 9, 12} × τ ∈ {1, .5, .3, .2, .1} + (1, 0.05); churn @1024
462M — τ alone	CoBit-M unguided, step 250k (of a 500k cosine)	Karras	τ ∈ {1, .7, .5, .3, .2, .1, .05}; churn @1024
superseded	CoBit-S unguided 425k upload (EMA; raw-weight diagnostic)	Karras	the first Exp. 1 attempt; appendix

Differences from the note's protocol, and why

- **Additions:** a w=12 column (the earlier GSM8K CFG sweep was still rising at w=7–20; the note's “flat” sweep was the Sudoku one) and a 512-step control. The γ=0 check was run for every τ.

- **Schedules.** Exp. 2 and CoBit-M use Karras; Exp. 1 uses the schedule its anchor picked. Every comparison is paired within one setting against its own $\tau=1$ cell, so the conclusions hold, but absolute accuracies are only comparable within an arm.
- **One seed.** Intervals are bootstrap over problems; paired deltas use the same problems and seed.
- **Checkpoint.** Exp. 1 runs on the 500k `last.pt` from the shared Drive folder (global_step 500000, raw + EMA weights, SHA-256 65cedca93bbd7c1c...), the same run as the uploaded 425k, 75k steps later.
- **w=1** runs through the two-branch CFG path (algebraically the conditional model).

2 τ alone



Each point is one cell (1319 problems); bars are bootstrap 95% CIs. Schedules differ between arms (legend), so compare shapes, not heights. In the shaded band the churn step is unstable (Section 5) and every cell scores 0%.

CoBit-S unguided, step 500k — churn, 1024 steps (entropic)

τ	accuracy %	95% CI	Δ vs τ=1, paired (95% CI)	programs that ran %
1	28.1	[25.8, 30.6]	—	63.5
0.7	32.4	[29.9, 35.0]	+4.3 [+2.4, +6.2]	69.2
0.5	2.7	[1.8, 3.6]	-25.5 [-27.9, -23.1]	5.0
0.3	0.0	[0.0, 0.0]	-28.1 [-30.6, -25.8]	0.0
0.2	0.0	[0.0, 0.0]	-28.1 [-30.6, -25.8]	0.0
0.1	0.0	[0.0, 0.0]	-28.1 [-30.6, -25.8]	0.0
0.05	0.0	[0.0, 0.0]	-28.1 [-30.6, -25.8]	0.0

CoBit-M 462M, unguided, step 250k — churn, 1024 steps (Karras)

τ	accuracy %	95% CI	Δ vs τ=1, paired (95% CI)	programs that ran %
1	31.5	[29.1, 34.0]	—	67.4
0.7	35.5	[32.9, 38.1]	+3.9 [+2.0, +5.9]	74.6
0.5	38.4	[35.8, 40.9]	+6.8 [+4.7, +8.9]	77.3
0.3	39.3	[36.7, 41.8]	+7.7 [+5.5, +9.9]	78.5
0.2	38.1	[35.5, 40.8]	+6.6 [+4.5, +8.8]	78.9
0.1	0.0	[0.0, 0.0]	-31.5 [-34.0, -29.1]	0.0
0.05	0.0	[0.0, 0.0]	-31.5 [-34.0, -29.1]	0.0

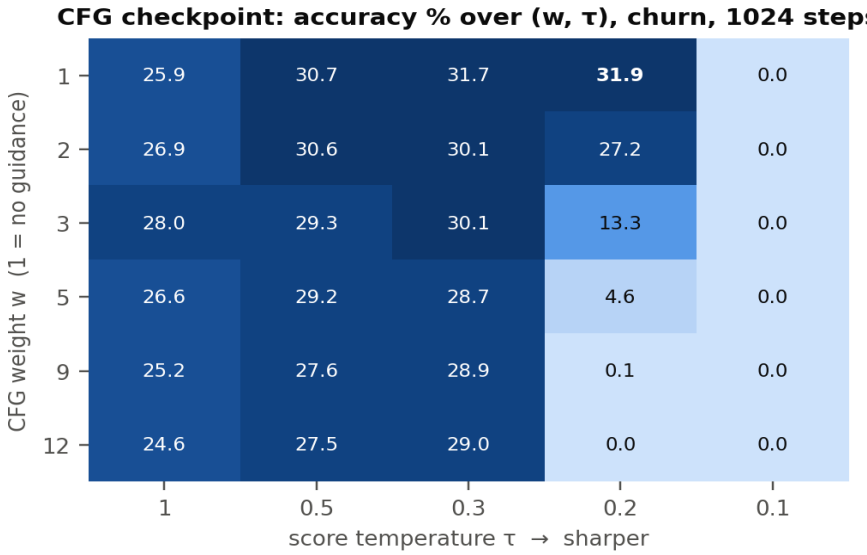
CoBit-S CFG checkpoint at w=1 (no guidance), step 425k — churn, 1024 steps (Karras)

τ	accuracy %	95% CI	Δ vs τ=1, paired (95% CI)	programs that ran %
1	25.9	[23.6, 28.3]	—	60.5
0.5	30.7	[28.3, 33.2]	+4.8 [+2.7, +6.8]	69.7
0.3	31.7	[29.2, 34.2]	+5.8 [+3.8, +7.8]	72.6
0.2	31.9	[29.4, 34.5]	+6.0 [+3.9, +8.1]	75.2
0.1	0.0	[0.0, 0.0]	-25.9 [-28.3, -23.6]	0.0
0.05	0.0	[0.0, 0.0]	-25.9 [-28.3, -23.6]	0.0

Observations

- **The gain is real where sampling is stable.** For CoBit-M, the CFG checkpoint, every τ tested in 0.2–0.7 beats τ=1 with a paired CI excluding zero. For the 500k unguided model at least one τ in that range does not (see its table).
- **Programs run more often when sharpened** (CoBit-M 67% → 78%): most of the gain is fewer broken programs.
- **Model size.** CoBit-M gains +7.7 pts and the CFG checkpoint +6.0; the 500k unguided CoBit-S +4.3. These are different checkpoints (scale, optimiser, conditioning dropout, schedule), not one model at several quality levels.
- **Where sampling breaks most likely depends on the σ grid, not the model.** On Karras (CoBit-M, CFG checkpoint) the churn step fails below τ≈0.15. On the entropic grid (500k unguided) the final step, σ 0.157 → 0.002, covers 99% of the remaining σ at once and fails below τ≈0.5 (Section 5), so that curve stops at 0.7.

3 CFG × τ on the CFG checkpoint



Accuracy % per cell, churn, 1024 steps, Karras, 1319 problems. w=1 is the conditional model. Bold = best cell.

- **τ does the work; w does not add to it.** Best cell: w=1, τ=0.2 (31.9%). CFG alone peaks at w=3 (28.0%, +2.0 [+0.1, +4.0] pts over w=1). Every w>1 row at its best τ is at or below the w=1 row at its best τ.
- **Guidance moves the breaking point.** At τ=0.2 accuracy falls from 31.9% (w=1) to 13.3% (w=3) and ≈0 for w≥9: CFG and τ both enlarge the drift, so their combination has to stay inside the stable range.
- **Sigma's settings:** (2, 0.1) → 0.0%, (9, 1) → 25.2%, (1, 0.05) → 0.0%. Their optimum lies in the region our sampler cannot reach.
- **Where the branches disagree.** The per-step CFG direction $\|D_c - D_u\|$ (rms over free bits) is ≈0.18 at σ>10 for every w, 0.12–0.20 at σ∈[1,10], ≤0.09 at σ∈[0.1,1] and 0 below: guidance acts mostly while the sample is still choosing its mode. A weak unconditional branch (10% dropout on a 130M model) is a plausible reason CFG adds little.

4 Experiment 1 on the unguided CoBit-S model

Schedule anchor (500k EMA, τ=1, churn, 1024 steps)

schedule	accuracy %	95% CI	programs that ran %
entropic	28.1	[25.8, 30.6]	63.5
Karras	26.0	[23.7, 28.4]	59.3

500k EMA — deterministic (γ=0), 1024 steps (entropic)

τ	accuracy %	95% CI	Δ vs τ=1, paired (95% CI)	programs that ran %
1	12.2	[10.5, 14.0]	—	31.5
0.7	24.7	[22.4, 27.1]	+12.5 [+10.3, +14.6]	63.1
0.5	4.4	[3.3, 5.5]	-7.8 [-9.8, -5.8]	47.2
0.3	0.3	[0.1, 0.6]	-11.9 [-13.7, -10.2]	12.7
0.2	0.4	[0.1, 0.8]	-11.8 [-13.6, -10.1]	23.6
0.1	0.9	[0.5, 1.4]	-11.3 [-13.1, -9.6]	40.8
0.05	1.1	[0.6, 1.7]	-11.1 [-13.0, -9.3]	52.6

500k EMA — churn, 512 steps (entropic): step-count control

τ	accuracy %	95% CI	Δ vs τ=1, paired (95% CI)	programs that ran %
1	26.8	[24.4, 29.2]	—	60.0
0.7	31.5	[29.0, 34.0]	+4.8 [+2.7, +6.9]	68.8
0.5	2.2	[1.4, 3.0]	-24.6 [-27.0, -22.1]	5.5
0.3	0.0	[0.0, 0.0]	-26.8 [-29.2, -24.4]	0.0
0.2	0.0	[0.0, 0.0]	-26.8 [-29.2, -24.4]	0.0
0.1	0.0	[0.0, 0.0]	-26.8 [-29.2, -24.4]	0.0
0.05	0.0	[0.0, 0.0]	-26.8 [-29.2, -24.4]	0.0

512 vs 1024 steps differ by at most 1.4 pts at any τ≥0.2, so the τ gain is not a step-size correction.

What went wrong with the first attempt (425k upload)

The uploaded 425k (SHA-256 9a6817c7c4871104…) scored 1.7% at τ=1 with churn and 11.6% deterministic under Karras. The first version of this report blamed a stale EMA. A CPU check disproves that: the EMA holds every trained tensor (only the fixed `time_fn._freq` buffer is absent, as in the healthy CFG checkpoint), is finite, and moved with the weights between 350k and 425k (median relative change 31% for EMA vs 33% for raw; EMA-to-raw distance median 8.9%). The anchor above rules out the σ schedule as well: the 500k file works under both. The cause is specific to the uploaded 425k file (the copy in the shared Drive folder is byte-identical to it (same SHA-256), so this is the Drive artifact itself; the June handoff records that the 29.42% came from a main-run 425k that was never saved to the shared folder) and is not identified. Its raw weights did score higher than its EMA under Karras (18.3% vs 4.9% at 512 steps), consistent with the observation that running weights work better for this model.

5 Where sampling breaks: the step size, not the readout

With churn γ=0.41 each step first raises σ to $\hat{\sigma} = 1.41\sigma$, then takes an Euler step to the next σ. The step contracts the distance to the denoiser output by a factor $1 - r/\tau$ with $r = 1 - \sigma_{next}/\hat{\sigma} \in [0.29, 0.31]$ on a 1024-step grid (median 0.30). It is stable only while $|1 - r/\tau| < 1$, i.e. $\tau > r/2 \approx 0.15$. At τ=0.2 the factor is ≈-0.5 (overshooting but stable); at τ=0.1 it is ≈-2.0, the state grows geometrically, and the diagnostics show it going non-finite by step 67 (σ≈78), before any token has formed. All 17 churn cells at τ≤0.1 score 0% with no program running.

The entropic grid has one much larger step. CoBit's entropic schedule (used for the 500k arm) ends at σ=0.157 and then jumps to σ=0.002 in a single step, r=0.987, while every other step has r≤0.13 (≈0.30 with churn). For that final step the same condition gives $\tau > 0.49$, with or without churn: at τ=0.5 the step overshoots the denoiser output by ~2× just before readout and flips bits. This matches the 500k collapse at τ≤0.5 in both its churn and deterministic tables, while Karras (largest deterministic step r=0.024) lets the other arms go down to τ=0.2. Leaving the final step unsharpened — Sigma reads out from an unsharpened call — removes it: Section 7 confirms this on the real model.

So the note's proposed reading (“if the executable rate collapses at $\tau \gtrsim 0.3$, the bitstream readout is the limit”) cannot be applied to churn cells: the collapse is numerical. Sigma samples with DDPM ancestral steps, which have no churn jump; whether CoBit stays stable at τ≤0.1 under such a step is untested (the rolling sampler's η=1 bridge is the natural place to test it).

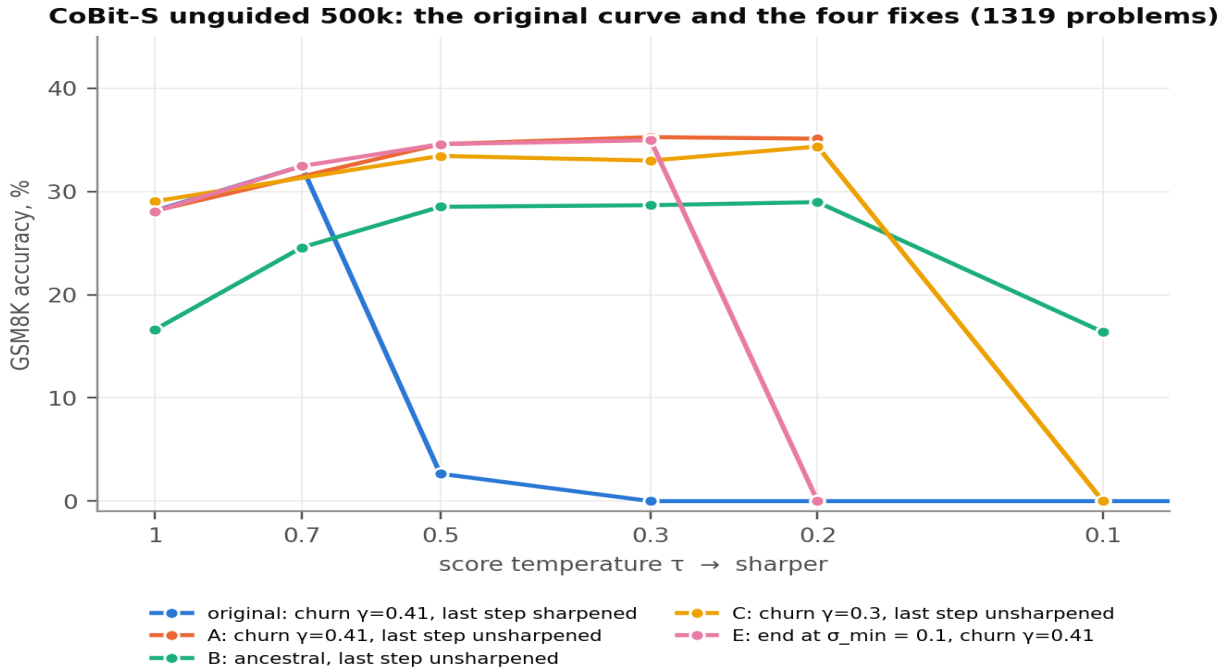
6 What this does and does not show

- **Does:** constant score temperature over the whole trajectory — the operation our earlier A1 sweep did not test — at τ≈0.2–0.3 with churn, on 2 of 3 checkpoints tested with paired CIs excluding zero across that range. The earlier A1 negative (late-σ only) does not transfer to this operation.
- **Does not:** reach Sigma's regime (τ≤0.1, w≥2); test CFG fairly (our unconditional branch comes from 10% dropout on a 130M model, Sigma's from attention masking at 3B); or separate model quality from model identity (one checkpoint per model, one seed).
- **Earlier related result:** the FKC drift-only β sweep (K=1, no weights) was constant scaling for τ≥0.77 and showed +4 pts at τ=0.83; consistent with Section 2.

The fixes proposed here (unsharpened last step, ancestral sampler, lower γ, logit temperature, ending at σ_min = 0.1) were run on 13 Oct: Section 7.

7 Fixing the collapse below τ = 0.7 (runs of 13 Oct)

All cells: CoBit-S unguided 500k EMA, entropic grid, 1024 steps, 1319 problems, seed 0, paired with the cells of Section 4. A: the last step (onto σ_min, the readout) uses the untempered drift. B: the VE ancestral sampler (exact Gaussian bridge, no churn). C: lower churn γ, last step unsharpened. D: DeepMind's logit temperature T (softmax over the valid codewords with the full logit, every step). E: the trajectory ends at σ_min = 0.1 instead of 0.002 (see entropy_profile.pdf).



Accuracy per cell (each 1319 problems). A's τ=1 point is the original τ=1 cell (the two settings coincide at τ=1).

A — unsharpened last step

sampler	τ	last step sharpened: acc / run / invalid	unsharpened: acc / run / invalid	unsharpened Δ vs τ=1 (paired)
churn γ=0.41	0.5	2.7 / 5.0 / 1.39	34.6 / 72.6 / 0.00	+6.4 [+4.4, +8.5]
churn γ=0.41	0.3	0.0 / 0.0 / 40.40	35.3 / 75.4 / 0.00	+7.1 [+5.1, +9.2]
churn γ=0.41	0.2	0.0 / 0.0 / 41.06	35.1 / 74.0 / 0.00	+7.0 [+4.9, +9.1]
deterministic	0.5	4.4 / 47.2 / 0.00	4.4 / 47.2 / 0.00	-7.8 [-9.8, -5.8]
deterministic	0.3	0.3 / 12.7 / 0.00	0.3 / 12.7 / 0.00	-11.9 [-13.7, -10.2]
deterministic	0.2	0.4 / 23.6 / 0.00	0.4 / 23.6 / 0.00	-11.8 [-13.6, -10.1]

B — ancestral sampler (no churn), last step unsharpened

τ	accuracy %	95% CI	Δ vs τ=1, paired (95% CI)	programs that ran %
1	16.6	[14.6, 18.7]	—	41.8
0.7	24.6	[22.3, 26.9]	+8.0 [+6.1, +9.9]	59.0
0.5	28.5	[26.1, 30.9]	+11.9 [+9.9, +14.1]	66.6
0.3	28.7	[26.2, 31.1]	+12.1 [+9.9, +14.3]	69.1
0.2	29.0	[26.5, 31.4]	+12.4 [+10.0, +14.8]	71.9
0.1	16.4	[14.5, 18.4]	-0.2 [-2.7, +2.2]	78.8

C — churn γ × τ, last step unsharpened (accuracy %)

γ \ τ	1	0.5	0.3	0.2	0.1	best Δ vs own τ=1
0.1	26.8	33.0	33.6	32.4	32.0	τ=0.3: +6.7 [+4.5, +9.0]
0.2	28.0	33.3	33.4	33.5	33.3	τ=0.2: +5.5 [+3.3, +7.7]
0.3	29.0	33.4	33.0	34.3	0.0	τ=0.2: +5.3 [+3.1, +7.4]
0.41 (orig.)	28.1	2.7	0.0	0.0	0.0	sharpened last step

E — end the trajectory at σ_min = 0.1

end point / last step	τ	accuracy %	programs run %	invalid tokens %	Δ vs τ=1 at σ_min 0.1
σ_min 0.1, sharpened	1	28.1	63.5	0.00	—
σ_min 0.1, sharpened	0.7	32.4	69.2	0.00	+4.4 [+2.5, +6.2]
σ_min 0.1, sharpened	0.5	34.6	72.6	0.00	+6.5 [+4.5, +8.6]
σ_min 0.1, sharpened	0.3	35.0	74.9	0.00	+6.9 [+4.9, +8.9]
σ_min 0.1, sharpened	0.2	0.0	0.0	13.60	-28.1 [-30.5, -25.7]
σ_min 0.1, unsharpened	0.3	35.3	75.4	0.00	+7.2 [+5.2, +9.2]
σ_min 0.1, unsharpened	0.2	35.1	74.0	0.00	+7.1 [+4.9, +9.2]

D — DeepMind logit temperature (churn γ=0.41)

logit temperature T	accuracy %	programs run %	invalid tokens %	Δ vs churn τ=1 (paired)
0.5	28.1	61.6	0.00	-0.1 [-1.9, +1.7]
0.3	28.5	64.3	0.00	+0.4 [-1.4, +2.1]
0.1	29.5	61.5	0.00	+1.4 [-0.6, +3.3]

Observations

- **The churn collapse was the last step.** With it unsharpened, churn γ=0.41 keeps working to τ=0.2 (34.6 / 35.3 / 35.1% at τ=0.5 / 0.3 / 0.2, against 2.7 / 0.0 / 0.0% before) and invalid tokens fall back to 0. Best: τ=0.3, +7.1 [+5.1, +9.2] pts over τ=1.
- **The deterministic collapse is not.** Unsharpening the last step leaves it unchanged (4.4% at τ=0.5): it is the mode/length collapse of Section 5, as predicted.
- **Ancestral** is stable to τ=0.2 (29.0%, +12.4 [+10.0, +14.8] pts over its own τ=1) but starts much lower (16.6% at τ=1, little noise) and drops back at τ=0.1 (16.4%). It never beats churn.
- **Lower γ extends the range**: at τ=0.1, γ=0.1 / 0.2 score 32.0 / 33.3% while γ=0.3 and 0.41 diverge, as the stability rule predicts. But no γ beats γ=0.41 with an unsharpened last step at its best τ; lower γ buys range, not accuracy.
- **Ending at σ_min = 0.1** alone recovers τ=0.5–0.3 (34.6 / 35.0%) but a sharpened last step still corrupts bits at τ=0.2 (0.0%). With the last step unsharpened, ending at 0.1 or 0.002 gives identical answers: once that step is untempered, it decides nothing.
- **DeepMind's logit temperature** is stable at every T (no invalid tokens), but helps far less: best T=0.1, 29.5% (+1.4 [-0.6, +3.3] pts over churn τ=1).

One seed throughout; the best cells should get seeds 1–2 before the numbers are quoted externally.

Appendix — superseded first attempt: unguided CoBit-S 425k upload (Karras, EMA)

churn, 1024 steps

τ	accuracy %	95% CI	Δ vs τ=1, paired (95% CI)	programs that ran %
1	1.7	[1.1, 2.5]	—	5.7
0.7	0.8	[0.3, 1.3]	-1.0 [-1.8, -0.2]	2.4
0.5	0.3	[0.1, 0.6]	-1.4 [-2.2, -0.8]	0.7
0.3	0.1	[0.0, 0.2]	-1.7 [-2.4, -1.0]	0.2
0.2	0.0	[0.0, 0.0]	-1.7 [-2.5, -1.1]	0.2
0.1	0.0	[0.0, 0.0]	-1.7 [-2.5, -1.1]	0.0
0.05	0.0	[0.0, 0.0]	-1.7 [-2.5, -1.1]	0.0

churn, 512 steps

τ	accuracy %	95% CI	Δ vs τ=1, paired (95% CI)	programs that ran %
1	4.9	[3.8, 6.1]	—	14.1
0.7	2.9	[2.0, 3.8]	-2.0 [-3.4, -0.8]	7.8
0.5	0.2	[0.0, 0.4]	-4.8 [-6.0, -3.6]	0.9
0.3	0.1	[0.0, 0.2]	-4.9 [-6.1, -3.7]	0.4
0.2	0.1	[0.0, 0.2]	-4.9 [-6.1, -3.7]	0.2
0.1	0.0	[0.0, 0.0]	-4.9 [-6.1, -3.8]	0.0
0.05	0.0	[0.0, 0.0]	-4.9 [-6.1, -3.8]	0.0

deterministic (γ=0), 1024 steps

τ	accuracy %	95% CI	Δ vs τ=1, paired (95% CI)	programs that ran %
1	11.6	[9.9, 13.3]	—	29.7
0.7	23.7	[21.5, 26.0]	+12.1 [+10.0, +14.3]	58.9
0.5	10.8	[9.2, 12.5]	-0.8 [-3.0, +1.4]	56.6
0.3	0.6	[0.2, 1.1]	-11.0 [-12.8, -9.2]	18.5
0.2	0.2	[0.0, 0.5]	-11.4 [-13.1, -9.7]	19.0
0.1	0.2	[0.0, 0.5]	-11.4 [-13.1, -9.7]	28.4
0.05	0.5	[0.2, 0.8]	-11.1 [-13.0, -9.4]	20.6

Raw-weight diagnostic, churn, 512 steps: τ=1 → 18.3%, τ=0.3 → 5.6%.

Data: 80 result files in results/scoretemp/eval_gh200 (+2 in anchor500/, +2 in diag/; the 14 silently-deterministic first-run cells are kept in invalid_s425_churn_silently_deterministic/ and unused). Sweep driver scripts/hpc/arch/scoretemp_gh200.sh; code snapshot logs/arch/scoretemp_provenance/. Built by docs/pdf_reports/build_scoretemp.py.